Adapting Native Geometry Decoders for Dynamic Actor Surfaces: Interim Evidence on Coverage and Observed Free Space

WorldSim V7.3 research draft

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Abstract

We study metric surface reconstruction of rigid dynamic Actors from sparse LiDAR and multi-camera observations, with known identities, calibration and trajectories. The prototype adapts a pretrained native geometry decoder and uses spatial queries to generate explicit local surfaces in Actor coordinates. An interim development study shows that better coverage does not ensure better physical first intersections: the completed joint candidate R5 increases surface recall and reduces missing returns, but introduces more early surfaces and observed-free-space intrusion than strong same-input references. Changing the static background reduces a separate source of intrusion while exposing a coverage cost. These observations motivate controlled comparisons of label span, geometry adaptation and surface-based objectives. The report records completed experiments and their limitations. A fully trained native decoder control improves fit-set hit rate but lowers development hit in all five logs. Its matched joint comparison, the next objective comparison, and independent new-log confirmation remain unfinished. No final method superiority is claimed.

1 Task and research hypothesis

For Actor i, we infer a reusable surface $\mathcal{S}_i$ in metric canonical coordinates from a fixed build window. Known rigid motion places it at query time: $\mathcal{S}_i(t) = T_{w\leftarrow i}(t)\mathcal{S}_i$. Camera calibration and trajectories are read-only. We focus on rigid vehicle categories; learned motion prediction, nonrigid deformation, intensity, complete no-return beam simulation and photorealistic rendering are outside this experiment.

The hypothesis is that adapting a pretrained geometric decoding path and jointly reading local multi-view and LiDAR evidence can improve surface position and coverage without increasing violations of observed free space. This is an empirical hypothesis, not a consequence of adding parameters or attention. Earlier small-adapter results did not directly test it: their final surface displacement was computed from a physical hidden state, while current visual features affected an evidence update. In that interface, displacement had zero direct derivative with respect to current visual features when parameters and other inputs were fixed. V7.3 opens a trainable visual-to-surface path; a negative outcome for one implementation cannot reject all geometric foundation-model adaptation.

2 Implemented model and supervision

2.1 Native decoding and spatial queries

The official VGGT training framework already supports freezing the aggregator while fine-tuning task heads [1]. Our prototype trains the full native DPT depth head (32,654,562 parameters), reading four aggregation levels. Only the completely frozen prefix is cached. No upper-aggregation LoRA is trained in the present joint candidate. Direct native-depth supervision uses build Actor LiDAR projected to the correct camera time, with the original metric scale fixed by robust build-side alignment.

The query decoder has 1,670,517 parameters, hidden width 192 and three update layers. Up to 1,024 build-evidence queries and 512 completion queries carry canonical positions and learned states. Completion is seeded by aligned native geometry, with a LiDAR fallback; the explicit image-only entry also permits learned coarse positions when both point sources are absent. The older main runs did not train that last

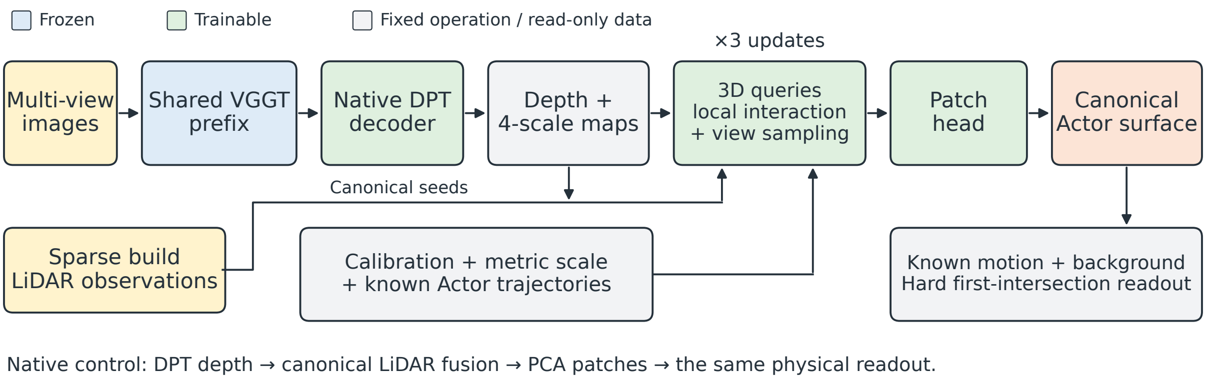

Figure 1: Implemented reconstruction components. Blue marks the frozen shared prefix; green marks trainable DPT, query and patch modules. Gray denotes fixed operations, supplied geometry or intermediate representations, not a gradient barrier: visual features and native seed positions can influence the generated physical surface. Calibration, metric scale and trajectories remain read-only. The native control replaces the query/patch pathway with canonical fusion and PCA patches, retaining the physical first-intersection readout. Supervision uses fit measurements; additional fit targets are not prediction inputs. The diagram does not imply that all variants or objectives have established a benefit.

input regime. For a query x_i, view v is read at

$$x_{c,v} = T_{c\leftarrow w}(t_v)T_{w\leftarrow i}(t_v)x_i. \tag{1}$$

Each query samples four local offsets at each of four feature scales, retaining camera, time and direction information. This follows the reference-centered sparse sampling idea of Deformable DETR [6], transferred here to calibrated Actor queries. Query chunks have size 128. Local 16-neighbor interactions use a spatial index with detached neighbor identities; we do not construct a dense query–query distance matrix. This reduces the interaction support to local neighborhoods; nearest-surface supervision still costs $O(PF)$ for P target points and F faces, with chunked temporary arrays. The prototype's image mask represents calibrated frustum and valid-image support, not verified occlusion visibility.

Each query predicts an explicit 3×3 patch with eight triangles, normal and bounded local bending, on a fixed 0.06 m grid spacing. Query centers are not limited to small displacements around original LiDAR candidates. Adjacent vertices within a patch are connected, but different patches are not welded into a closed manifold. Appearance opacity cannot hide a physical surface.

2.2 One surface, distinct measurement semantics

The implemented objectives act on this explicit geometry:

$$L = L_{\text{coverage}} + \lambda_n L_{\text{native}} + \lambda_f L_{\text{free}} + 0.05 L_{\text{envelope}} + \lambda_e L_{\text{event}}. \tag{2}$$

Coverage is the mean distance from sampled measured target points to the nearest triangle. Sparse targets do not define a complete surface: an unmeasured region is not automatically empty, and we do not penalize all generated points merely for lacking a nearby sparse target. Native supervision is axial-depth SmoothL1 with a 0.2 m transition. It remains available at known build correspondences even when current predicted geometry leaves the Actor support region. This addresses a previously observed self-closing native-support path.

For the original beam with first measured range d_r^* and predicted first intersection $\hat{d}_r$, the hard free-space objective is

$$L_{\text{free}} = \frac{1}{|R|} \sum_{r \in R} \mathbf{1}[\hat{d}_r < \infty] \, [d_r^* - 0.2 - \hat{d}_r]_+. \tag{3}$$

All near-box original beams contribute; an occluded rear Actor does not replace the original first-return distance. Missing predictions have zero free penalty and remain penalized through measured coverage and reported missing-return rates. Regions behind the original return remain unknown. A finite beam-tube version uses the same surface with fixed width 0.03 m and 32×32 sampling; it does not learn confidence or transparency.

The event term is a capped, fixed-footprint likelihood of the same first visible surface with Gaussian range width 0.2 m and cap 28. It is evaluated on the owned subset of the same sampled beams. Repeating a surface does not add an independent termination opportunity. Depth-termination supervision has prior art in DS-NeRF [4]; changing an expected-depth loss to a likelihood is not claimed as novel. Crucially, an entirely absent surface receives the cap and zero position gradient from this term. Coverage and free-space geometry must retain their separate roles. A local differentiable visibility boundary is not a guarantee of global support discovery or convergence. The rasterized visibility and antialiasing derivatives use nvdiffrast [7].

3 Data, comparators and estimands

The nuScenes development study contains 489 rigid Actor entries in 31 windows from 25 independent logs: 414 fit entries in 20 logs and 75 development entries in five logs. The main runs admit 371 fit and 67 development entries with build LiDAR; the remaining 43 and eight stay in the population with explicit missing predictions. Each window supplies six cameras at four build times, jointly encoded at 378×672. Two interleaved times supply evaluation returns. Development logs never supply optimizer updates to the shared models.

Additional full-track labels are fit-only: 414 target files contain 1,690,284 measured Actor points, including positive targets for some empty-build entries. Those labels never become prediction inputs. R5 and R6 use the short-window label regime; R7–R9 use the longer fit-only labels. R11 has completed the native full-track control; its matched joint counterpart R10 is still running. R12 isolates the beam-tube free objective relative to R10, and the running native R14 is its same-objective control. An equal-capacity pointwise control has not been trained on this full population; therefore a spatial-attention causal claim is currently unsupported.

We retain LiDAR accumulation with PCA patches, native adapted geometry plus canonical LiDAR fusion, native TSDF fusion, a CAPA-style build-only adaptation and a trained AdaPoinTr reference. CAPA supplies an official sparse-measurement test-time adaptation formulation [3]; our transfer resets its low-rank parameters per window, takes 100 build-only updates and fuses its outputs in canonical coordinates. Its per-image affine alignment and test-time budget differ from shared training. AdaPoinTr [2] uses the official PCN initialization, 30 fit epochs, axis conversion and full-track labels. It has a different optimizer, pretraining corpus and output density. These are meaningful competitors, not all single-factor ablations.

An owned original return is a hit within ± 0.2 m, early below that interval, late above it, or missing if no surface intersects. These outcomes include misses in the denominator. Free-space intrusion uses all original near-box beams. Measured surface recall is the fraction of heldout Actor measurements within 0.2 m of the explicit surface. Frames are weighted by their actual measurements within each Actor, followed by Actor means within logs and equal log weighting. Paired uncertainty resamples independent logs 10,000 times, not individual rays. With only five development logs, intervals remain limited evidence. Twenty-three development Actors have no owned heldout return; they are retained and their unavailable metrics are identified.

4 Completed development evidence

R5 completes 30 effective epochs (11,130 updates) of genuine native DPT and query adaptation. The final recall is 79.71%, but early returns reach 20.98% and mean free intrusion reaches 0.3517 m. Relative to its own initialization, missing returns decrease while early returns increase. Relative to LiDAR R6, the surface is closer to heldout measurements but physical first intersections are worse (Table 2). These are not contradictory metrics: additional support can cover true measurements and still lie in front of other observations. The main candidate has not established a physical advantage.

A separate observed-return point evaluation uses only original owned beam intersections and measured targets, with missing predictions retained. R5's F-score is 0.3541, compared with R6's 0.4319 and R9's 0.4922. R5 minus R6 is -0.0778 with log-bootstrap interval $[-0.1173, -0.0382]$. This is an observable subset, not full canonical-surface precision. Unsquared two-direction Chamfer is conditional on nonempty predicted/measured sets and is never substituted for the missing-rate column. Nearest-neighbor matching can also hide a wrong-beam intersection; literal first-return metrics remain necessary.

The R7–R9 sequence isolates narrower mechanisms on LiDAR queries. Longer fit labels reduce early/free

Method	Hit (%)	Early (%)	Miss (%)	Free (m)	Recall (%)
LiDAR PCA	21.24	4.35	73.41	0.0177	65.42
Actor TSDF + carving	3.28	0.26	96.33	0.0008	21.51
Native fusion	22.78	9.16	57.35	0.1503	78.13
LiDAR R6 (window labels)	31.01	14.02	49.31	0.0838	72.42
LiDAR R7 (track labels)	31.75	11.84	49.90	0.0716	74.55
LiDAR R8 (beam free)	31.00	5.59	56.83	0.0343	71.95
LiDAR R9 (+event)	32.47	6.01	54.57	0.0385	72.47
CAPA-style TTA R2	17.80	10.95	58.46	0.1203	76.90
AdaPoinTr R2	14.29	8.54	54.24	0.1221	84.60
Joint R5 (window labels)	22.28	20.98	37.71	0.3517	79.71
Native DPT R11 (track labels)	17.61	7.64	60.09	0.1650	75.15

Table 1: Completed candidates, all 75 development entries in five logs. Recall is measured-target-to-surface recall, not complete-surface recall. Empty predictions remain missing. Short/long labels, adaptation budgets and output representations differ as described in the text; no row is a final confirmed method. Late rates are the residual outcome and are omitted for space.

Joint R5 minus	Metric	Difference	95% log bootstrap
R5 initialization	Early (pp)	+8.706	[+1.221, +18.493]
R5 initialization	Miss (pp)	-13.433	[-16.301, -8.843]
LiDAR R6	Hit (pp)	-8.728	[-14.097, -4.478]
LiDAR R6	Free (m)	+0.268	[+0.056, +0.503]
Native fusion	Early (pp)	+11.826	[+5.400, +23.016]
Native fusion	Free (m)	+0.201	[+0.037, +0.400]

Table 2: Selected paired development differences, five independent logs. Positive early/free and negative hit are adverse. R5 versus R6 shares the short-window label regime, but differs in visual input and native adaptation; it does not isolate attention. The recovered R5 run also has the random-state limitation described in Section 6.

relative to R6. The beam-tube free objective then reduces early/free further while lowering measured recall. Adding event supervision reduces missing returns relative to R8, but the hit/free paired intervals span zero; event support is still absent on roughly 39% of sampled owned training beams. None of these results proves that event likelihood alone recovers missing geometry.

Query-source accounting attributes 91.74% of R5's log-weighted free intrusion and 78.37% of its early-return contribution to completion-initialized patches. Their centers average 1.087 m from the nearest original build point. This measures source provenance, not causal visual attribution or displacement from each query's initial position: both source groups pass through the same trainable spatial/visual updates.

4.1 Completed native control: fit improvement does not transfer

R11 trains the entire native DPT for 30 epochs with full-track fit labels, the original hard-range free objective and fixed PCA surface construction. There are 11,130 Actor presentations and 10,550 actual optimizer updates. The 580 skipped presentations have no effective native gradient: 420 have no camera pose and 160 have no native support and fall back to fixed LiDAR. All lack native build-depth correspondences. The query decoder is frozen. The largest recorded native projection-weight change from M1 initialization is 0.005224; this is genuine native adaptation, not a frozen-output adapter.

The measurement-weighted native Huber loss drops from 0.8592 to 0.3619 m, and fit-log hit increases by 2.28 points, with interval $[0.93, 3.70]$. Those fit measurements are in the training-label span. On the five development logs, hit decreases from 22.78% to 17.61%, declining in every log; early decreases from 9.16% to 7.64%. Mean free intrusion changes from 0.1503 to 0.1650 m, without a clear improvement. The paired evidence is shown in Table 3.

The old native-fusion reference generated surfaces for five additional zero-build-LiDAR development entries that R11 retains as missing. Its conditional distance mean therefore has a different denominator. Initial R11 and native fusion coincide on the 67 build-ready entries; comparisons of distance use common evaluable entries rather than subtracting unmatched means. R11 also trails same-full-track LiDAR R7 in hit by 14.14

Metric	Difference	95% log bootstrap	Improved logs
Hit (pp)	-5.166	[-10.920, -1.125]	0/5
Early (pp)	-1.512	[-2.872, -0.660]	5/5
Miss (pp)	+2.741	[-2.697, +8.178]	2/5
Free (m)	+0.015	[-0.007, +0.031]	1/5
One-way distance (m)	+0.006	[-0.007, +0.028]	4/5
Recall (pp)	-2.979	[-8.339, +1.135]	3/5

Table 3: Native R11 final minus its own initialization on five development logs, with existing 10,000-resample log-bootstrap intervals (seed 7304). Improvement means higher hit/recall or lower early/miss/free/distance. The one-way distance is conditional on common evaluable support.

points, interval $[-18.58, -8.62]$. Comparison with R9 additionally changes beam/event objectives.

This configuration has not improved development physical reconstruction. It does not identify whether the cause is observation contamination, shared parameter drift, discrete support/readout, or objective semantics. LoRA3D's scene adaptation [8] and CAPA's sparse-measurement adaptation [3] motivate possible restricted or build-side adaptation controls; neither establishes this failure's cause. We first retain the matched R10/R11 and R12/R14 comparisons. Rejecting all visual geometric adaptation, or repeating the already evaluated CAPA setup unchanged, would not follow from this result.

5 Density, static background and composition

Point-cloud conversion changes the physical readout. AdaPoinTr R2 retains all 16,384 generated points, but its main comparison uses a common limited set of patch centers. Reading PCA patches at all native output points increases development hit by 24.72 percentage points and early by 30.85 points, while increasing free intrusion by 0.2190 m. A denser surface is not an unqualified improvement. Native Actor TSDF at 0.1 m voxels and 0.3 m truncation produces a nonempty mesh for only 162 of 489 entries (18 of 75 development entries). Its low free intrusion therefore coexists with 96.33% missing returns. This is an applicability limit of the stated sparse configuration, not a general negative result about TSDF fusion.

At scene level we preserve background/Actor ownership and cast against

$$\mathcal{S}_{\text{scene}}(t) = \mathcal{S}_{\text{static}} \cup \bigcup_i T_{w\leftarrow i}(t)\mathcal{S}_i. \tag{4}$$

The closest actual intersection across all owners is the scene first return. All methods share background inputs, Actor trajectories and the readout. Static construction excludes all known boxes at each build observation time. Original build beams provide one-pass face carving before the measured return; this does not fill unobserved background behind Actors.

Using VDBFusion [5] with fixed 0.1 m voxels, 0.3 m truncation and no hole filling yields a substantially different background support. On the five-log nuScenes development background-only comparison, TSDF plus carving reduces mean free intrusion from 0.2656 to 0.0822 m, but increases missing returns from 53.28% to 62.80%. Every log improves free intrusion and loses hit coverage. With this common TSDF background, R5 still exceeds R9 in Actor cohort early returns by 15.33 points and free intrusion by 0.1084 m; both paired intervals exclude zero. Background correction does not remove the Actor geometry problem.

One already-exposed AV2 development log provides a separate implementation check. Its points are already compensated into reference-ego coordinates; endpoints are transformed to world once, while each return retains its actual time-dependent sensor origin. The single-origin VDBFusion API is called in groups of exactly equal origins, with no time/origin averaging. Relative to carved PCA background, carved TSDF reduces background-only free intrusion from 0.5130 to 0.1075 m but increases missing from 32.13% to 45.54%. Changing only the background leaves R5's moving-Actor outcomes unchanged on its 88 moving cohort beams. This single development log cannot supply between-log uncertainty or independent confirmation.

6 Remaining experiments and claim boundaries

R10 still requires completed training and analysis. R12 is registered but unrun; R14 has begun native-only training with the same 0.03 m, 32^2 beam-tube free objective registered for R12. Both start from the original M1 initialization with seed 7304; neither resumes the hard-free candidate. The objective comparison keeps the old cohort and all view/time inputs.

R13 has completed one epoch over all 41 camera-supported fit entries among the 51 original zero-build-LiDAR entries, from fresh M1 initialization. All 41 updates produce positive DPT and query gradient norms. There are no build native-depth correspondences in this cohort: visual decoder gradients flow through generated geometry and sampled features. Six training presentations use the coarse support fallback. Of seven entries without a positive full-track target, two have an active free penalty and five satisfy the sampled free constraints and update only through the box-envelope term. Therefore not all 41 updates are sensor-residual-driven. Unknown regions are not negative targets.

The final surfaces cover 41 of 43 fit entries and five of eight development entries; the five entries without camera poses remain missing. The sole owned development return changes from missing to an early intersection, with range error 3.4622 m. Its target-to-surface distance is only 0.0119 m, illustrating that a nearby surface can coexist with an incorrect earlier surface. We do not infer accuracy or a meaningful uncertainty interval from one return. This one-epoch diagnostic establishes execution, not sufficient fitting or generalization. Full training of the expanded cohort remains a separate experiment from the free-objective comparison.

Twenty additional AV2 logs were selected by available build metadata before model quality evaluation. Their 936 Actor entries, raw beams and 28-view frozen prefixes are prepared. Their TSDF backgrounds are also complete: 6,839,969 original background returns yield 6,962,322 raw triangles; one-pass build carving removes 146,991 (2.11%), leaving 6,815,331. Zero remaining build violations reflect the construction constraint, not heldout correctness. No model-quality confirmation has been run on these logs. The final method and background choice must precede that readout. This is cross-dataset confirmation at a different camera/resolution budget, not an i.i.d. nuScenes test. Project-unseen logs also do not prove absence from foundation-model pretraining.

Current joint and native-only configurations run on the RTX 3090 while preserving the stated time/view inputs. The joint peak allocated memory is approximately 10.2 GiB; native-only training uses approximately 2.35 GiB. Twenty-eight-view AV2 prefix extraction at 672^2 peaks at 7.64 GiB. These are operation-specific allocated peaks, not a sum or a bound for deeper PEFT. Container RAM is capped at 90 GiB. Concurrent run wall time is not exclusive GPU time.

The original R5 process disappeared after epoch 21 without an available traceback; the cause remains unknown. Recovery restored model and optimizer but its legacy checkpoint lacked RNG state. The recovered run is not a bitwise continuation: 107 incomplete-epoch updates were discarded, and the observed combined wall-time lower bound is 9.32 hours. The run provenance remains part of its interpretation.

Annotation boxes with margins approximate point ownership and overlap regions are excluded; they are not dense instance ground truth. nuScenes here lacks per-return timestamps, whereas the AV2 adapter has per-return sensor origins with a documented, development-inferred dual-LiDAR ID convention. Frustum support is not occlusion truth. The learned patches are not globally manifold. Trajectory-edit examples demonstrate composition only: with no counterfactual ground truth, released unknown background is not a correctness measurement. These limitations, the small development-log count and the unfinished controls preclude a final reconstruction or sensor-simulation superiority claim.

Evidence and reproducibility

The source branch is `research/worldsim-v7.3-geometric-adaptation`, derived directly from the final V7.2 state. The current operational authority is `docs/RESEARCH_STATUS.md`; `docs/RESEARCH_FAILURES.md` and `docs/EXPERIMENTS.md` retain failures and experiment history. Tables in this draft are generated from saved JSON summaries, without rerunning inference or selecting observations. Raw run directories under `/root/autodl-tmp/runs/worldsim_v73/` retain configurations, checkpoints, training logs and explicit surfaces. `paper_v73/README.md` identifies the exact compact evidence files and compilation commands. This draft is not an arXiv submission and does not mark V7.3 as complete.

References

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